

Mandatory Carbon Disclosure and the Cost of Capital

Ingram, Alice · Okonkwo, Ngozi

Fictional example paper — generated by Lectern for demonstration.

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Abstract

We exploit a staggered national rollout of mandatory carbon disclosure to estimate its effect on the cost of equity capital. Firms brought into scope report a decline in implied cost of capital of roughly 38 basis points relative to not-yet-treated peers, concentrated among firms with previously sparse voluntary reporting. The effect is larger where an assurance requirement accompanies the mandate, and absent where disclosure is unaudited. We interpret this as evidence that the credibility of the disclosure, not its existence, is what investors price.

1. Introduction

Whether mandatory sustainability disclosure changes the cost of capital is an old question with a thin empirical record. Most settings introduce a mandate everywhere at once, leaving no clean counterfactual. We use a rollout that phased firms in by size threshold over four years, which lets us compare firms brought into scope against firms not yet in scope but subject to the same regime.

Our contribution is twofold. First, we provide a credible causal estimate in a setting with staggered adoption. Second, we show that the effect depends on whether the mandate carries an assurance requirement, which speaks directly to the debate over whether disclosure regulation should specify verification.

2. Institutional setting

The regime obliges in-scope firms to publish scope 1 and scope 2 emissions in the annual report, with scope 3 required only where material. Phase-in was determined by a headcount threshold measured two years before the reporting year, which firms could not plausibly manipulate at the margin we study.

Assurance was required only for firms above a second, higher threshold. This gives us a within-mandate comparison between assured and unassured disclosers, holding the disclosure requirement itself constant.

3. Results

Table 3 reports the baseline difference-in-differences estimate. The coefficient on the treatment indicator is -0.38 percentage points with a standard error of 0.11, clustered by firm. Event-study estimates show no differential pre-trend in the three years before phase-in.

Splitting by assurance, the effect for assured disclosers is -0.61 percentage points, while the estimate for unassured disclosers is statistically indistinguishable from zero. We take this as the central result of the paper.